

LAPORAN PRAKTIKUM FRAMEWORK WEB MODUL 12



Disusun oleh :

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MODUL 12

(WEB SERVIC RESET API)

Langkah- langkah web service user register dan login:

1. Api Authentication using Laravel Sactum

Program:

```
C:\Users\Lenovo\Downloads\SEMESTER 5\PRAKTIK FW\prak_crud>composer require laravel/sanctum
Cannot use laravel/sanctum's latest version v4.2.1 as it requires php ^8.2 which is not satisfied by your platform.
./composer.json has been updated
Running composer update laravel/sanctum
Loading composer repositories with package information
Updating dependencies
Nothing to modify in lock file
Writing lock file
Installing dependencies from lock file (including require-dev)
Nothing to install, update or remove
Generating optimized autoload files
Class Database\Seeders\KotaSeeder located in ./database/seeders/kotaSeeder.php does not comply with psr-4 autoloading standard (rule: Database\Seeders\ => ./database/seeders). Skipping.
> Illuminate\Foundation\ComposerScripts::postAutoloadDump
> @php artisan package:discover --ansi

 INFO  Discovering packages.

laravel/breeze ..... DONE
laravel/sail ..... DONE
laravel/sanctum ..... DONE
laravel/tinker ..... DONE
nesbot/carbon ..... DONE
nunomaduro/collision ..... DONE
nunomaduro/termwind ..... DONE
pestphp/pest-plugin-laravel ..... DONE
spatie/laravel-ignition ..... DONE

86 packages you are using are looking for funding.
Use the `composer fund` command to find out more!
> @php artisan vendor:publish --tag=laravel-assets --ansi --force

 INFO  No publishable resources for tag [laravel-assets].

Found 1 security vulnerability advisory affecting 1 package.
```

2. Install api dengan : Php artisan install:api
3. Tambahkan di model user:

Program:

```
User.php X create.blade.php ...kota X index.php create.blade.php ...provinsi index.blade.php
app > Models > User.php > User > casts
1  <?php
2  namespace App\Models;
3  // use Illuminate\Contracts\Auth\MustVerifyEmail;
4  use Illuminate\Database\Eloquent\Factories\HasFactory;
5  use Illuminate\Foundation\Auth\User as Authenticatable;
6  use Illuminate\Notifications\Notifiable;
7  use Laravel\Sanctum\HasApiTokens;
8  class User extends Authenticatable{
9      use HasApiTokens, HasFactory, Notifiable;
10     /**
11      * The attributes that are mass assignable.
12      *
13      * @var array<int, string>
14      */
15     0 references
16     protected $fillable = [
17         'name',
18         'email',
19         'password',
20     ];
21
22     /**
23      * The attributes that should be hidden for serialization.
24      *
25      * @var array<int, string>
26      */
27     0 references
28     protected $hidden = [
29         'password',
30         'remember_token',
31     ];
32
33     /**
34      * The attributes that should be cast.
35      *
36      * @var array<string, string>
37      */
38     0 references
39     protected $casts = [
40         'email_verified_at' => 'datetime',
41         'password' => 'hashed',
42     ];
43 }
```

4. Jalankan migrasi sanctum dengan: `php artisan migrate`

Program: tampil nothing karna tabel personal access tokens sudah ada di database

```
C:\Users\Lenovo\Downloads\SEMESTER 5\PRAKTIK FW\prak_crud>php artisan migrate
INFO Nothing to migrate.
```

5. Membuat API untuk register dengan laravel-sanctum

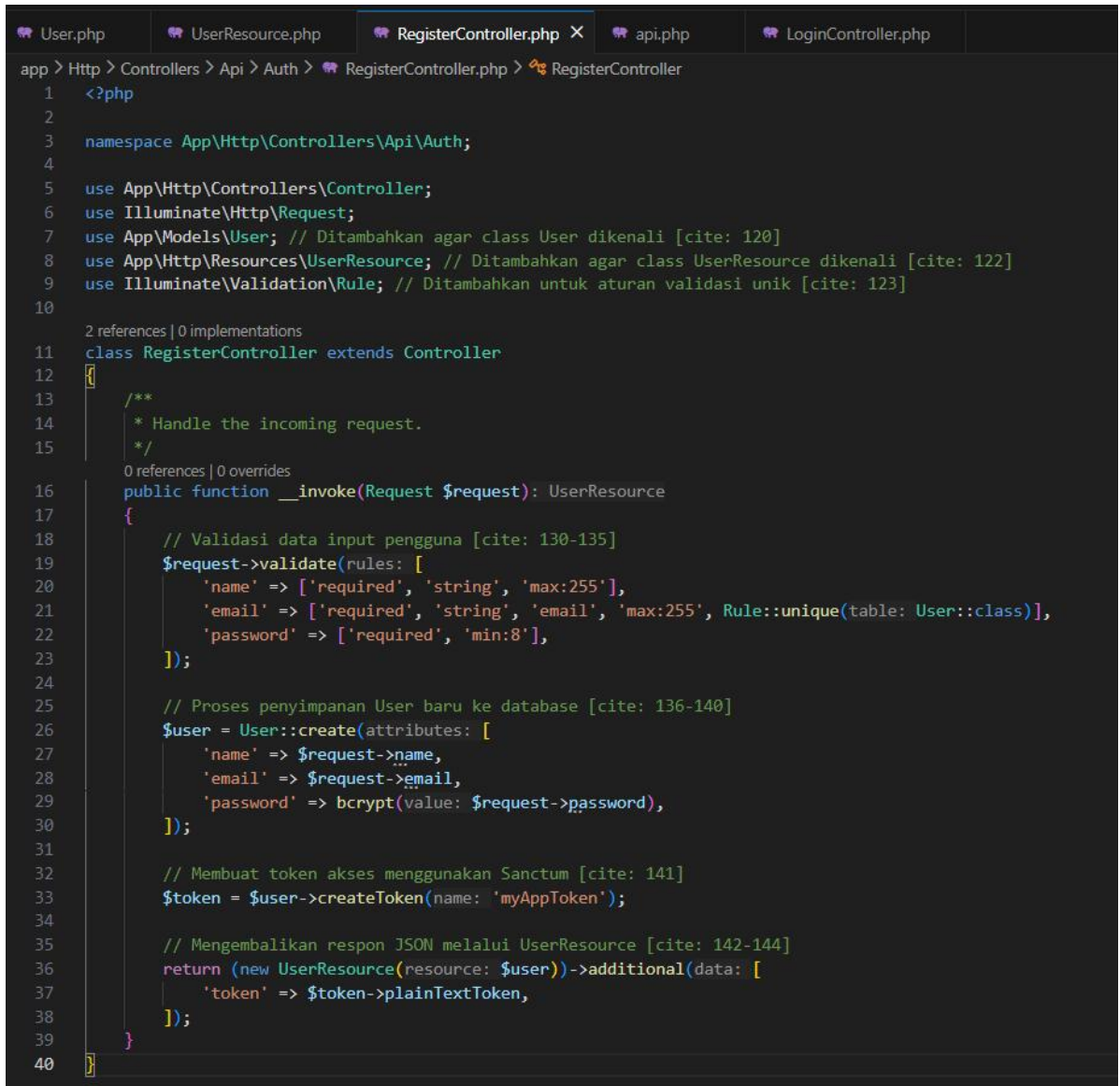
6. Gunakan perintah berikut:

`php artisan make:controller Api\Auth\RegisterController -i`

program:

```
C:\Users\Lenovo\Downloads\SEMESTER 5\PRAKTIK FW\prak_crud>php artisan make:controller Api/Auth/RegisterController -i
```

INFO Controller [C:\Users\Lenovo\Downloads\SEMESTER 5\PRAKTIK FW\prak_crud\app\Http\Controllers\Api\Auth\RegisterController.php] created successfully.



```
app > Http > Controllers > Api > Auth > RegisterController.php > RegisterController
1  <?php
2
3  namespace App\Http\Controllers\Api\Auth;
4
5  use App\Http\Controllers\Controller;
6  use Illuminate\Http\Request;
7  use App\Models\User; // Ditambahkan agar class User dikenali [cite: 120]
8  use App\Http\Resources\UserResource; // Ditambahkan agar class UserResource dikenali [cite: 122]
9  use Illuminate\Validation\Rule; // Ditambahkan untuk aturan validasi unik [cite: 123]
10
11  2 references | 0 implementations
12  class RegisterController extends Controller
13  {
14      /**
15       * Handle the incoming request.
16       */
17      0 references | 0 overrides
18      public function __invoke(Request $request): UserResource
19      {
20          // Validasi data input pengguna [cite: 130-135]
21          $request->validate(rules: [
22              'name' => ['required', 'string', 'max:255'],
23              'email' => ['required', 'string', 'email', 'max:255', Rule::unique(table: User::class)],
24              'password' => ['required', 'min:8'],
25          ]);
26
27          // Proses penyimpanan User baru ke database [cite: 136-140]
28          $user = User::create(attributes: [
29              'name' => $request->name,
30              'email' => $request->email,
31              'password' => bcrypt(value: $request->password),
32          ]);
33
34          // Membuat token akses menggunakan Sanctum [cite: 141]
35          $token = $user->createToken(name: 'myAppToken');
36
37          // Mengembalikan respon JSON melalui UserResource [cite: 142-144]
38          return (new UserResource(resource: $user))->additional(data: [
39              'token' => $token->plainTextToken,
40          ]);
41      }
42  }
```

7. Kemudian gunakan perintah : `php artisan make:resource UserResource`

Program:

```
C:\Users\Lenovo\Downloads\SEMESTER 5\PRAKTIK FW\prak_crud>php artisan make:resource UserResource
```

INFO Resource [C:\Users\Lenovo\Downloads\SEMESTER 5\PRAKTIK FW\prak_crud\app\Http\Resources\UserResource.php] created successfully.

```
User.php  UserResource.php X RegisteredUserController.php RegisterController.php api.php
app > Http > Resources > UserResource.php > UserResource > toArray()
1  <?php
2
3  namespace App\Http\Resources;
4
5  use Illuminate\Http\Request;
6  use Illuminate\Http\Resources\Json\JsonResource;
7
8  4 references | 0 implementations
9  class UserResource extends JsonResource
10 {
11     /**
12      * Transform the resource into an array.
13      *
14      * @return array<string, mixed>
15      */
16     0 references | 0 overrides
17     public function toArray($request): array
18     {
19         return [
20             'id' => $this->id,      // [cite: 111]
21             'name' => $this->name,  // [cite: 112]
22             'email' => $this->email, // [cite: 113]
23         ];
24     }
25 }
```

8. Selanjutnya daftarkan pada router/api.php

Program:

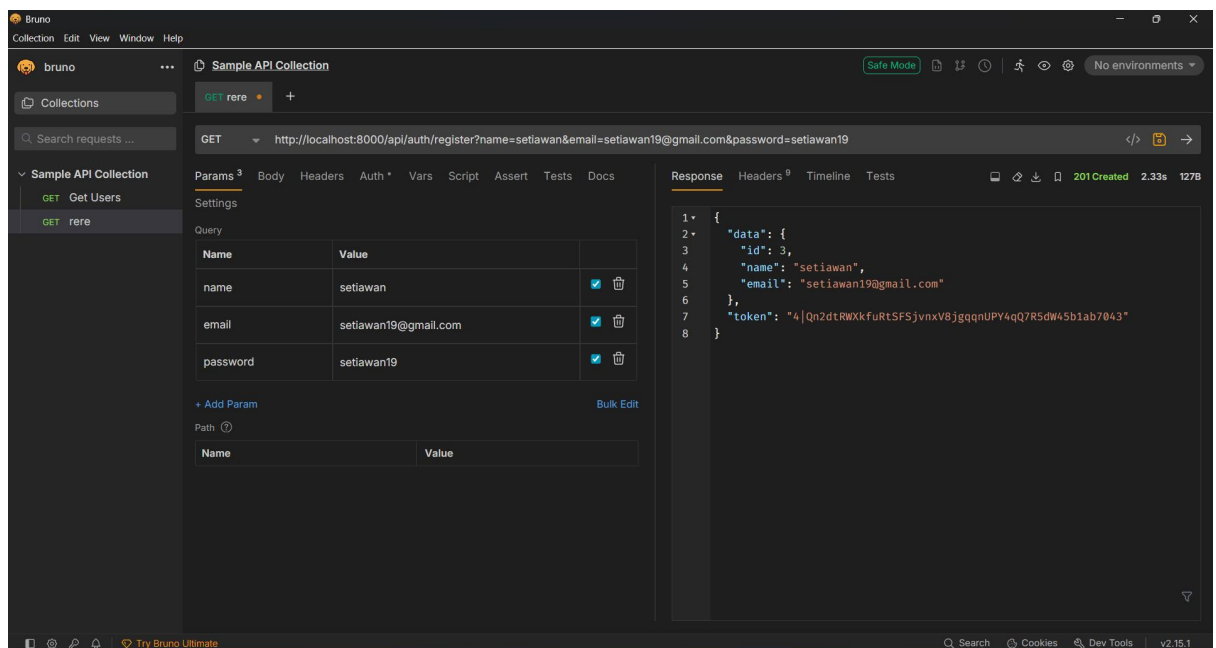
```
User.php  UserResource.php RegisteredUserController.php api.php X LoginController.php
routes > api.php
1  <?php
2
3  use Illuminate\Http\Request;
4  use Illuminate\Support\Facades\Route;
5  use App\Http\Controllers\Api\Auth\RegisterController;
6  use App\Http\Controllers\Api\Auth>LoginController;
7
8  // Route untuk Register [cite: 148-149]
9  Route::get(uri: 'auth/register', action: RegisterController::class);
10
11 // Route untuk Login [cite: 260]
12 Route::get(uri: 'auth/login', action: \App\Http\Controllers\Api\Auth>LoginController::class);
13 /*
14 |-----
15 | API Routes
16 |-----
17 |
18 | Here is where you can register API routes for your application. These
19 | routes are loaded by the RouteServiceProvider and all of them will
20 | be assigned to the "api" middleware group. Make something great!
21 |
22 */
23
24 Route::middleware(middleware: 'auth:sanctum')->get(uri: '/user', action: function (Request $request): mixed {
25     return $request->user();
26 });
27
```

9. Kemudian lakukan pengujian dengan posman disini saya menggunakan bruno

Program:

```
User.php UserResource.php RegisterController.php X api.php LoginController.php
app > Http > Controllers > Api > Auth > RegisterController.php > RegisterController
1 <?php
2
3 namespace App\Http\Controllers\Api\Auth;
4
5 use App\Http\Controllers\Controller;
6 use Illuminate\Http\Request;
7 use App\Models\User; // Ditambahkan agar class User dikenali [cite: 120]
8 use App\Http\Resources\UserResource; // Ditambahkan agar class UserResource dikenali [cite: 122]
9 use Illuminate\Validation\Rule; // Ditambahkan untuk aturan validasi unik [cite: 123]
10
11 class RegisterController extends Controller
12 {
13     /**
14      * Handle the incoming request.
15      */
16     public function __invoke(Request $request): UserResource
17     {
18         // Validasi data input pengguna [cite: 130-135]
19         $request->validate(rules: [
20             'name' => ['required', 'string', 'max:255'],
21             'email' => ['required', 'string', 'email', 'max:255', Rule::unique(table: User::class)],
22             'password' => ['required', 'min:8'],
23         ]);
24
25         // Proses penyimpanan User baru ke database [cite: 136-140]
26         $user = User::create(attributes: [
27             'name' => $request->name,
28             'email' => $request->email,
29             'password' => bcrypt(value: $request->password),
30         ]);
31
32         // Membuat token akses menggunakan Sanctum [cite: 141]
33         $token = $user->createToken(name: 'myAppToken');
34
35         // Mengembalikan respon JSON melalui UserResource [cite: 142-144]
36         return (new UserResource(resource: $user))->additional(data: [
37             'token' => $token->plainTextToken,
38         ]);
39     }
40 }
```

Tampilan register:



Program:

```
User.php  UserResource.php  RegisterController.php  api.php  LoginController.php X
app > Http > Controllers > Api > Auth > LoginController.php > LoginController
1  <?php
2
3  namespace App\Http\Controllers\Api\Auth;
4
5  use App\Http\Controllers\Controller;
6  use Illuminate\Http\Request;
7  use App\Http\Resources\UserResource; // PENTING: Harus ditambahkan agar tidak error
8
9  2 references | 0 implementations
10 class LoginController extends Controller
11 {
12     /**
13      * Handle the incoming request.
14      */
15     0 references | 0 overrides
16     public function __invoke(Request $request): JsonResponse|UserResource
17     {
18         // Validasi input email dan password
19         $credentials = $request->validate(rules: [
20             'email' => ['required'],
21             'password' => ['required'],
22         ]);
23
24         // Proses pengecekan kredensial di database
25         if (auth()->attempt(credentials: $credentials)) {
26             $user = auth()->user(); //
27
28             // Mengirim respon menggunakan UserResource dan menyertakan token
29             return (new UserResource(resource: $user))->additional(data: [
30                 'token' => $user->createToken(name: 'myAppToken')->plainTextToken,
31             ]);
32         }
33
34         // Respon jika email atau password salah
35         return response()->json(data: [
36             'message' => 'Your credential does not match.',
37         ], status: 401);
38     }
39 }
```

Tampilan login:

